

Program 7: Deploy a web application to Kubernetes using Minikube

Step 1: Start the minikube and check status

```
1rv24mc040-haripriya@haripriya-ASUS:~$ minikube start
🐳 minikube v1.37.0 on Ubuntu 24.04
🌟 Using the docker driver based on existing profile
👉 Starting "minikube" primary control-plane node in "minikube" cluster
📡 Pulling base image v0.0.48 ...
🔄 Restarting existing docker container for "minikube" ...
🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
🔍 Verifying Kubernetes components...
   ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟 Enabled addons: default-storageclass, storage-provisioner

! /usr/local/bin/kubectl is version 1.31.0, which may have incompatibilities with Kubernetes 1.34.0.
   ▪ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
🏠 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

Step 2: Check status of minikube

```
1rv24mc040-haripriya@haripriya-ASUS:~$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

Step 3: Create a nginx-deployment file

nano nginx-deployment.yaml

```
GNU nano 7.2 nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
```

Step 4: Apply the yaml created

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program7$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment created
```

Step 5: Check for the deployments and pods

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program7$ kubectl get deployment
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
nginx-deployment    0/2     2            0           17s
```

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program7$ kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
kubia                              1/1    Running             1 (18m ago) 3d23h
nginx-deployment-6f9664446b-7bzg4  0/1    ImagePullBackOff    0           26s
nginx-deployment-6f9664446b-r695t  0/1    ContainerCreating   0           26s
```

Step 6: Expose the deployment

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program7$ kubectl expose deployment nginx-deployment
\
> --type=NodePort \
> --port=80
service/nginx-deployment exposed
```

Step 7: Get the services to check for deployments

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program7$ kubectl get services
NAME                TYPE        CLUSTER-IP    EXTERNAL-IP  PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1     <none>       443/TCP          3d23h
kubia               NodePort    10.101.170.73 <none>       8080:30668/TCP   3d23h
nginx-deployment    NodePort    10.100.98.171 <none>       80:32012/TCP     13s
```

Step 8: Describe svc

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program7$ kubectl describe svc nginx-deployment
Name:                nginx-deployment
Namespace:           default
Labels:              <none>
Annotations:         <none>
Selector:            app=nginx
Type:               NodePort
IP Family Policy:    SingleStack
IP Families:         IPv4
IP:                 10.100.98.171
IPs:                10.100.98.171
Port:               <unset> 80/TCP
TargetPort:         80/TCP
NodePort:           <unset> 32012/TCP
Endpoints:          10.244.0.7:80,10.244.0.6:80
Session Affinity:    None
External Traffic Policy: Cluster
Internal Traffic Policy: Cluster
Events:             <none>
```

Step 9: Get the nginx official page by using the below command in the browser

```
1rv24mc040-haripriya@haripriya-ASUS:~/Devops/program7$ minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:32012

🌐 Opening service default/nginx-deployment in default browser...

